L	Topic
1	Axiomatic Definition of Probability, Sum Rule, Product Rule
2	Independent Events and Conditional Probability
3	Bayes’ Theorem
4	Tutorial / Problems
5	One-Dimensional Random Variables (Discrete & Continuous)
6	CDF, Expectation, Variance, Moment Generating Function
7	Tutorial / Problems
8	Binomial and Poisson Random Variables
9	Problems on Binomial and Poisson Distribution
10	Gamma, Exponential, and Chi-Square Distributions
11	Normal Distribution
12	Two-Dimensional Random Variables
13	Covariance and Correlation Coefficient
14	Tutorial / Problems
15	Functions of One-Dimensional Random Variables
16	Functions of Two-Dimensional Random Variables
17	Problems on Functions of Random Variables
18	Random Processes and Markov Chain
19	Transition Matrix
20	Tutorial / Problems
21	Sampling Theory and Central Limit Theorem
22	Point Estimation and Maximum Likelihood Estimator (MLE)
23	Interval Estimation
24	Tutorial / Problems
31	Optimization: Basic Solution, Convex Sets and Functions
32	Simplex Method
33	Tutorial / Problems
34	Optimization Using Gradient Descent
35	Constrained Optimization and Lagrange Multipliers
36	Tutorial / Problems